

FREDRIK PETTIT

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Education

University of British Columbia

Expected Graduation: May 2028

B.Sc. Computer Science (Artificial Intelligence) / GPA: 3.8

Vancouver, BC

Work Experience

AI Engineer Intern

Apr 2026 – Aug 2026

AEM

Langford, BC

- Scoped and shipped an AI customer support agent for customers in 4 countries that answers product questions straight from company docs at \$0.03 a ticket versus \$8 of staff time to save about **\$16K** a year while clearing an 800+ ticket backlog
- Engineered a RAG pipeline with hybrid search on Azure AI Search over 4,000 pages of bilingual docs and lifted auto-resolution from 34% to **44%** with query rewriting for the German and acronym-heavy queries behind most escalations
- Fine-tuned a PyTorch reranker on 2,000 past support tickets that lifted retrieval recall@5 from **77% to 87%**
- Built a 300-case regression eval that caught a 6-point drop in a GPT-4o to GPT-4.1 upgrade before it reached customers
- Cut monthly budget-vs-actuals reporting from **6+ hours to 5 minutes** with a LangChain agent that corrects vendor names and accounting codes, turns its confident answers into reusable rules and flags unsure ones for finance review

Software Engineering Intern

Jan 2026 – Apr 2026

ELLUSTROS

The Blue Mountains, ON

- Designed and deployed the FastAPI recommendation engine powering the company's React Native chair-fitting app, filtering out chairs that don't fit each user's body measurements and ranking the rest with weighted ergonomic scoring
- Cut photo upload failures on slow mobile networks by **70%** by sending images straight to S3 through presigned URLs
- Wrote 40+ Jest tests for sign-in and chair recommendation flows that stopped bugs from reaching QA

Teaching Assistant | CPSC 310, CPSC 110

Aug 2025 – Present

UBC Department of Computer Science

Vancouver, BC

- Supported 900+ students in UBC's largest computer science course, achieving a 4.9/5 average student evaluation score
- Wrote randomized JavaScript quiz questions and Python autograders used across CPSC 310 - Intro to Software Engineering

Projects

YouTube Shorts AI Optimizer | *Python, Claude API, MCP, Gemini, Google Cloud Run*

- Built an AI copilot that critiques draft YouTube Shorts before publishing and predicts where viewers will swipe away, helping grow my own channel from 1.2K to **38K** subscribers with **45M** views, and now used by **100+** other users
- Evaluated Gemini's retention predictions on 85 held-out Shorts at 0.45 Spearman, beating a veteran creator's 0.25
- Shipped a remote MCP server that returns a plan inside Claude in 18s at p95 by running tool calls in parallel
- Blocked **98.5%** of 200 prompt-injection attacks in viewer comments by isolating that text and making tools read-only

Apex FSD - StormHacks 2026 (Winner) | *Python, PyTorch, OpenCV, SDL2*

- Won Best Python Hack at Western Canada's largest hackathon with a self-driving AI that plays Forza Horizon by physically turning a real steering wheel and hands control back the instant a human grabs the wheel
- Trained a PyTorch imitation learning model with DAgger so every manual correction is recorded as new training data
- Used saliency maps to spot the model fixating on roadside signs and removed the crash footage that taught it

Scheming - HackMIT 2026 | *Python, Claude, OpenAI, Modal, MongoDB*

- Built an LLM lie detector for AI agents playing Mafia/Werewolf that flags the exact line where a player says one thing but privately plans another and matches human judgment on **94%** of 300 hand-labeled turns at a Cohen's kappa of 0.85
- Ran the same LLM-as-judge on 1,500 real coding-agent sessions and found claims of passing tests that the agent's own logs contradicted in 6% of runs at 92% precision, a check teams can run on a pull request before merging it

Technical Skills

Languages & Databases: Java, Python, C++, C#, SQL, R, JavaScript/TypeScript, PostgreSQL, DynamoDB, Redis
AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, OpenCV, sentence-transformers, embeddings, hybrid retrieval, RAG
LLM & Agents: LangGraph, LangChain, LangSmith, Claude/OpenAI/Gemini APIs, MCP, fine-tuning, guardrails, evals
Frameworks: Spring Boot, FastAPI, React, React Native, Expo, JUnit, Jest, Playwright, Pydantic, pandas, NumPy
Cloud & DevOps: AWS (Lambda, Step Functions, SQS), GCP Cloud Run, Azure AI Search, Docker, Git, CI/CD